

Problem-Solution Fit

Date	30 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

1. CUSTOMER SEGMENT(S) [CS] Meteorologists, Disaster Management Authorities, Relief Coordinators, Government Agencies, and Residents in Flood- Prone Areas.	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] Limited real-time data access, resource constraints, and internet availability in some regions.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Traditional weather reports and manual monitoring. Pros: Widely available. Cons: Time- consuming and less accurate.
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] Delayed flood warnings, inaccurate manual forecasting, and difficulty identifying high- risk regions during every monsoon season.	9. PROBLEM ROOT / CAUSE [RC]	7. BEHAVIOR + ITS INTENSITY [BE] Frequent monitoring during monsoon periods and high urgency during emergencies.
3. TRIGGERS TO ACT [TR] Heavy rainfall, rising river levels, severe weather alerts, and monsoon forecasts.	10. YOUR SOLUTION [SL] A machine learning-based flood prediction system using Decision Tree, Random Forest, KNN, and XGBoost integrated with a Flask web application.	8. CHANNELS OF BEHAVIOR [CH] ONLINE Web dashboard and cloud platform.
4. EMOTIONS BEFORE / AFTER [EM] Before: Concerned and uncertain. After: Informed and prepared.		OFFLINE Government alerts and emergency response teams.